



Fig. 2: OpenSimulator virtual simulation overview

and promoting healthy competition.

Educational and research opportunities. Open source metaverse platforms offer valuable educational and research opportunities. Students, researchers, and developers can study the code base, experiment with new ideas, and contribute to the advancement of virtual reality technologies.

Innovation and experimentation. Open source metaverse platforms often act as incubators for innovation and experimentation. The accessibility of the code base encourages developers to push boundaries, create new features, and explore novel use cases. This vibrant ecosystem drives rapid innovation and evolution within the metaverse space.

It's important to note that open source metaverse platforms also come with challenges, such as the need for technical expertise and potential fragmentation.

OpenSimulator

OpenSimulator (http://opensimulator.org/wiki/Main_Page), also termed as 'OpenSim', is an open source metaverse platform that allows individuals and organisations to create and deploy their own virtual worlds. It serves as a framework for building 3D virtual environments, offering a

rich set of features for user interaction, content creation, and collaborative experiences.

At its core, OpenSimulator enables users to design and customise their virtual worlds, constructing landscapes, buildings, and objects. Users can create and customise avatars, defining their appearance, clothing, and animations, fostering personal expression and individuality within the virtual environment. One of OpenSimulator's strengths lies in its support for real-time collaboration, enabling multiple users to interact and communicate with each other within the virtual space. It facilitates social interaction, teamwork, and the organisation of virtual events. OpenSimulator incorporates a scripting language called OSSL (OpenSimulator Scripting Language), enabling users to create interactive objects and automate behaviours within the virtual world. The platform includes a physics engine that simulates realistic physics interactions, such as gravity, collisions, and object dynamics, and enhances the realism and immersion of the virtual environment.

OpenSimulator supports the creation of multiple regions or simulators that can be interconnected to form a larger virtual world. Another

notable feature is the hypergrid connectivity, which allows users to teleport and explore other OpenSimulator-powered virtual worlds. It also supports content import and export, enabling users to import external assets, such as 3D models and textures, into their virtual worlds. There is an active community of users, developers, and content creators who contribute to the platform's development and provide support.

The key features of OpenSimulator are:

- Supports 3D virtual spaces of dynamic sizes
- Full support for multi-user cum online 3D environment
- Support for multiple clients and protocols
- Real-time physics simulation and supports inworld scripting using LSL/OSSL

JanusVR

JanusVR is an immersive web browsing platform that allows users to explore and interact with virtual reality (VR) environments. It combines the power of virtual reality with the accessibility of the web, providing a unique and immersive browsing experience.

It offers users seamless connectivity, allowing them to browse the